



American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

Animal Rescue and Aid Platform

A Software Engineering Project Submitted By

Semester: Summer_21_22		Section:	Group Number:	
SN	Student Name	Student ID	Contribution (CO3+CO4)	Individual Marks
1	MOHAMMED MOSTAFA	22-47170-1	16.66%	
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3	MD. MAHFUZUR RAHMAN	22-47078-1	16.66%	
4	NOWSHIN FARIHA	22-47074-1	16.66%	
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6	AFRIN BINTE AMIN	22-47146-1	16.66%	

The project will be evaluated for the following Course Outcomes

CO3: Select appropriate software engineering models, project management roles and their associated skills for the complex software engineering project and evaluate the sustainability of developed software, taking into consideration the societal and environmental aspects	Total Marks	
	Appropriate Process Model Selection and Argumentation with Evidence	[5 Marks]
	Evidence of Argumentation regarding process model selection	[5Marks]
	Analysis the impact of societal, health, safety, legal and cultural issues	[5Marks]
Submission, Defense, Completeness, Spelling, grammar and Organization of the Project report	[5Marks]	
CO4: Develop project management plan to manage software engineering projects following the principles of engineering management and economic decision process	Total Marks	
	Develop the project plan, its components of the proposed software products	[5Marks]
	Identify all the activities/tasks related to project management and categorize them within the WBS structure. Perform detailed effort estimation correspond with the WBS and schedule the activities with resources	[5Marks]
Identify all the potential risks in your project and prioritize them to overcome these risk factors.	[5Marks]	

Description of Student's Contribution in the Project work

Student Name: MOHAMMED MOSTAFA Student ID: 22-47170-1 Contribution in Percentage (%): 16.66% <u>Contribution in the Project:</u> ▪ Project Planning ▪ Research and Analysis _____ Signature of the Student
Student Name: SHARMIN AKHTER SAMIA Student ID: 22-46441-1 Contribution in Percentage (%):16.66% <u>Contribution in the Project:</u> ▪ Comparision with existing platforms ▪ References to the existing Platforms _____ Signature of the Student
Student Name: MD. MAHFUZUR RAHMAN Student ID: 22-47078-1 Contribution in Percentage (%):16.66% <u>Contribution in the Project:</u> ▪ Background to the problem ▪ Challenges in Animal rescue and welfare _____ Signature of the Student
Student Name: NOWSHIN FARIHA Student ID: 22-47074-1 Contribution in Percentage (%):16.66% <u>Contribution in the Project:</u> ▪ Categorizing our idea ▪ Solution to the problem _____ Signature of the Student

Student Name: ABDULLAH AL NOMAN

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Contribution in Percentage (%):16.66%

Contribution in the Project:

- Documentation Preparation
- Strategy Development

Signature of the Student

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Contribution in Percentage (%):16.66%

Contribution in the Project:

- Implementation Strategy
- Resource Allocation

Signature of the Student

1. PROJECT PROPOSAL

1.1 Background to the Problem

1.1.1 Global Stray Animal Crisis

The issue of stray animals is a global concern, impacting both developed and developing countries. Stray animals, including dogs, cats, and other species, face numerous challenges, such as lack of food, water, shelter, and medical care. This situation results in suffering for the animals and poses significant public health and safety risks. According to the World Health Organization, there are an estimated 200 million stray dogs worldwide. In the United States alone, approximately 3.3 million dogs and 3.2 million cats enter animal shelters each year. In developing countries, the problem is often more severe due to limited resources and infrastructure for animal control and welfare.

Stray animals are often carriers of diseases such as rabies, posing significant risks to human populations. Injured and sick animals on the streets can become aggressive, leading to increased incidents of attacks on humans. Additionally, uncontrolled breeding among stray animals leads to overpopulation, further exacerbating the problem.

1.1.2 Challenges in Animal Rescue and Welfare

The current systems for addressing the plight of stray and wounded animals are fragmented and often inadequate, presenting significant challenges in rescue, treatment, and adoption. Many animal rescue organizations and shelters operate with limited funding and resources, especially in rural and underprivileged areas, leading to insufficient veterinary services and delayed or inadequate medical care.

Public awareness and education also present challenges. The general public often lacks knowledge on how to assist stray animals or provide basic first aid. Additionally, misconceptions and fears about stray animals can prevent people from intervening.

Coordination and communication gaps further complicate rescue efforts. Poor coordination between rescue groups, shelters, and veterinary services leads to inefficiencies and duplicated efforts. The absence of a centralized platform for reporting and managing rescue cases hampers timely intervention.

Adoption efforts are also hindered by several factors:

- High rates of animal intake at shelters lead to overcrowding and increased euthanasia rates.
- Potential adopters often face difficulties in accessing information about animals available for adoption, which contributes to lower adoption rates.

1.1.3 Technological and Logistical Barriers

The underutilization of technology significantly impacts the effectiveness of animal rescue operations. There is limited use of data analytics and AI to identify and predict problem areas or to streamline rescue and treatment processes. Additionally, the absence of integrated platforms connecting all stakeholders—from rescuers and veterinarians to volunteers and adopters—further exacerbates these challenges.

Logistical issues also pose significant barriers. Inefficient transportation and logistics systems hinder the movement of injured animals to veterinary facilities, and the difficulty in tracking and managing the medical history and treatment of rescued animals complicates their care and recovery.

1.1.4 Categorizing Our Idea:

Our project falls into Category A based on the features and comparison with existing platforms. This categorization is defined by:

- **Innovative Use of Technology:** Incorporates AI-driven injury assessment and geolocation-based services that are not widely present in existing solutions.
- **Comprehensive Integration:** This method combines multiple functionalities (rescue operations, medical care, adoption, community engagement, education, and fundraising) into one cohesive platform.
- **Unique Offerings:** Features like real-time rescue tracking, AI-based treatment suggestions, and integrated crowdfunding for specific cases set it apart from other solutions.

1.1.4.1 Category A vs. Category B:

- **Category A:** Projects that offer novel, innovative solutions with a wide array of integrated features, utilizing advanced technologies and addressing multiple facets of the problem comprehensively.
- **Category B:** Projects that provide incremental improvements or focus on a single aspect of the problem, often using existing technologies in conventional ways.

Given the detailed and innovative nature of our proposed platform, with unique features and a holistic approach, it clearly aligns with Category A.

1.2 Solution to the Problem

The objective of this project is to develop a comprehensive platform that provides immediate rescue and medical care for stray and injured animals, enhances community involvement, ensures sustainable funding, and facilitates long-term welfare and adoption. The platform will utilize AI-driven technology for injury assessment, leveraging image recognition to evaluate the severity and type of injuries in animals. Based on this analysis, the AI will suggest appropriate first aid measures and direct users to the nearest veterinary services, ensuring timely care. Additionally, a geolocation-based directory will list nearby veterinary clinics, animal hospitals, and shelters, including emergency contact information, to streamline the rescue process.

The platform also aims to foster community involvement through volunteer coordination and educational resources. It will feature a volunteer management system where individuals can sign up, receive training, and participate in rescue operations or foster care. Educational materials, such as first aid tutorials and awareness campaigns, will be provided to promote animal welfare. Furthermore, community forums and success stories will encourage public engagement and support.

To ensure sustainable funding, the platform will integrate crowdfunding and donation options, enabling users to support specific rescue cases or contribute to general operations. Partnerships with NGOs, corporate sponsorships, and grant applications will be pursued to secure additional resources. The long-term welfare of rescued animals will be addressed through AI-based adoption and foster care matching, digital health records, and behavioral training, increasing the likelihood of successful adoptions. Lost and found services, including social media integration, will also be available to reunite lost pets with their owners.

The proposed solution employs state-of-the-art technology, including cloud services, microservices architecture, and a user-friendly interface, to provide a scalable and secure platform accessible via web and mobile applications. The technological infrastructure ensures that the platform is both effective and feasible in meeting business objectives, offering significant societal, health, safety, legal, and cultural impacts by addressing the issue of stray animals comprehensively.

The target group for this solution includes animal welfare organizations, volunteers, veterinarians, shelters, and the general public. These users will benefit from the platform's ability to coordinate rescue efforts, provide timely medical care, and facilitate adoptions. By offering tools and resources tailored to the needs of these groups, the platform will enhance the overall effectiveness of animal rescue operations.

In terms of contribution to scientific results, this project advances the application of AI in animal welfare by developing a novel system for injury assessment and treatment recommendations. The integration of AI with geolocation and real-time tracking technologies offers an innovative approach to solving the problem of stray animals, which has been well-documented but inadequately addressed in existing studies.

Several existing software solutions, such as Petfinder, Adopt-a-Pet.com, and RescueGroups.org, offer platforms for animal adoption, volunteer management, and fundraising. However, these solutions often lack comprehensive features for immediate rescue, medical care, and real-time tracking. Our proposed platform extends these existing solutions by integrating advanced AI-driven injury assessment, real-time updates, and a holistic approach to animal welfare, providing more benefits to users and ensuring a broader impact on animal rescue efforts.

In comparison with existing platforms, our solution offers unique features such as AI-based injury assessment, geolocation services for emergency care, real-time tracking of rescue operations, and a seamless adoption process enhanced by digital health records and behavioral training. This innovative approach addresses the limitations of current software solutions, making it a more effective tool for managing and improving animal rescue and welfare efforts.

1.2.1 Comparison with existing Platform:

Feature	Our Platform	Petfinder	Adopt-a-Pet.com	Pet Rescue Report	RescueGroups.org	ASPCA PTAS
AI-Driven Injury Assessment	Yes	No	No	No	No	No
Geolocation-Based Vet Directory	Yes	No	No	No	No	No
Real-Time Rescue Tracking	Yes	No	No	No	No	No
Volunteer Coordination	Yes	No	No	No	Yes	Yes
Crowdfunding and Donations	Yes	No	No	No	Yes	Yes
Educational Resources	Yes	Yes	Yes	No	No	Yes
Community Engagement	Yes	Yes	No	Yes	No	No
Digital Health Records	Yes	No	No	No	Yes	Yes
Adoption and Foster Care Matching	Yes	Yes	Yes	No	Yes	Yes
Mobile Application	Yes	Yes	Yes	No	No	Yes
Lost and Found Integration	Yes	No	No	Yes	No	No
Behavioral Assessment and Training	Yes	No	No	No	No	Yes

1.2.2 References to the existing platforms:

1. **Petfinder:**

- Website: [Petfinder](#)
- Features: Searchable database of adoptable pets, user profiles, adoption application processes, educational resources, and mobile app.

2. **Adopt-a-Pet.com:**

- Website: [Adopt-a-Pet.com](#)
- Features: Searchable listings of pets for adoption, tools for shelters and rescues, pet care information, and alerts for new pet listings.

3. **Pet Rescue Report:**

- Website: [Pet Rescue Report](#)
- Features: Reporting and searching for lost and found pets, news and articles on animal rescue, community engagement.

4. **RescueGroups.org:**

- Website: [RescueGroups.org](#)
- Features: Pet adoption portals, volunteer and foster management, fundraising and donation management, website, and data management tools.

5. **ASPCA's Pet Tracking and Adoption System (PTAS):**

- Website: [ASPCA](#)
- Features: Comprehensive animal care management, adoption and intake tracking, health and medical records management, reporting, and analytics.

2. SOFTWARE DEVELOPMENT LIFE CYCLE

2.1 Process Model

- The "Animal Rescue and Aid Platform" is a small-scale project aimed at creating a platform that connects rescuers, aid providers, and people seeking to adopt or assist animals in need. Given the project's nature, which involves dynamic requirements and the need for close collaboration among a small team of four to six members, it is essential to choose a development method that offers flexibility, iterative progress, and strong team collaboration. The **Scrum** methodology, an Agile model, is ideally suited for this environment due to its structured yet adaptable approach, making it the best fit for managing the development process effectively.
- Scrum stands out among other methodologies due to its ability to handle evolving requirements and its focus on iterative development. Unlike the Waterfall or V-model, which are more rigid and linear, Scrum allows for regular adjustments through time-boxed sprints, ensuring that the project can evolve based on feedback and changing needs. Compared to more complex models like the Spiral model, Scrum offers simplicity, making it easier for a small team to implement without the overhead of extensive risk management and documentation. Additionally, Scrum provides clear roles and ceremonies that promote regular communication and stakeholder involvement, which is crucial for the success of a collaborative project like this.
- The choice of Scrum is further supported by its proven effectiveness in similar environments. With its emphasis on iterative progress, regular stakeholder feedback, and a focus on team collaboration, Scrum minimizes risks and ensures that the project remains aligned with its goals throughout the development process. The structured framework of Scrum provides the necessary discipline to manage the project efficiently, while its flexibility allows the team to respond swiftly to any changes or new insights. This combination of structure and adaptability makes Scrum the most suitable methodology for developing the "Animal Rescue and Aid Platform," ensuring that the project is delivered successfully and meets the needs of its users.

2.2 Project Role Identification and Responsibilities

2.2.1 Roles/Stakeholders in Software Development:

- I. Product Owner
- II. Scrum Master
- III. Scrum Team
- IV. Customer
- V. Management

2.2.2 Responsibilities of the role in the software development:

I. Product Owner:

- **Requirement Definition:** The Product Owner is responsible for gathering and defining the project requirements from stakeholders, which involves understanding the needs of the users and translating them into features and functionalities.
- **Backlog Management:** The Product Owner maintains and prioritizes the Product Backlog, ensuring that the most valuable and important features are delivered first.
- **Stakeholder Communication:** This role serves as the main point of contact between the development team and stakeholders, ensuring that the team's work aligns with the project's objectives and stakeholder expectations.
- **Acceptance of Work:** The Product Owner is responsible for reviewing and accepting the work done by the development team at the end of each sprint, ensuring that it meets the defined criteria.

II. Scrum Master:

- **Facilitation:** The Scrum Master facilitates all Scrum ceremonies, including Sprint Planning, Daily Stand-ups, Sprint Reviews, and Retrospectives. They ensure that these meetings are productive and focused on achieving the sprint goals.
- **Impediment Removal:** The Scrum Master identifies and helps remove any impediments or obstacles that might hinder the team's progress, ensuring that the development process runs smoothly.
- **Process Guidance:** They guide the team in understanding and following Scrum practices and principles, helping to maintain the Scrum framework effectively.
- **Team Support:** The Scrum Master supports the development team by creating an environment that fosters collaboration, accountability, and continuous improvement.

III. Scrum Team:

- **Design and Development:** The development team is responsible for designing, coding, and testing the software. They work together to turn the Product Backlog items into a functional product increment by the end of each sprint.
- **Collaboration:** Team members collaborate closely, sharing knowledge and helping each other to solve problems and meet the sprint goals.
- **Quality Assurance:** The team ensures that the product meets the required quality standards through continuous testing, code reviews, and adherence to best practices.
- **Sprint Commitments:** The development team commits to completing a certain amount of work during each sprint and is accountable for delivering what they commit to.

IV. Customer:

- **Usage of the Software:** End users are the individuals who will ultimately use the "Animal Rescue and Aid Platform." Their needs and experiences are central to the design and development process.
- **Feedback:** End users provide feedback on the usability and functionality of the software, which helps in refining the product to better meet their needs.
- **Acceptance Testing:** In some cases, end users may be involved in acceptance testing to ensure that the software meets their requirements and is ready for deployment.

V. Management:

- **Requirement Input:** Stakeholders provide input on the requirements and expectations for the software, often representing the interests of specific user groups or organizations.
- **Feedback Provision:** They review the product increments presented at the end of each sprint and provide feedback that influences the direction of future development.
- **Project Approval:** Stakeholders are involved in key decision-making processes and may be responsible for approving major project milestones or the final product.

Rubric for Project Assessment (CO3)

Criteria	Marks distribution (Max 3X5= 15)				Acquired Marks
	Inadequate (1-2)	Satisfactory (3)	Good (4)	Excellent (5)	
Selection of Software Engineering Models	Does not articulate a position or argument of choosing appropriate model. Does not present any evidence to support the arguments for the choice of the model	Articulates a position or argument for choosing models that is unfocused or ambiguous. Presents incomplete/vague evidence to support argument for model choice	Articulates a position or argument of choosing models that is limited in scope. Does not present enough evidence to support the argument for the choice of the model	Clearly articulates a position or argument for the choosing software engineering models. Presents sufficient amount of evidence to support argument for the model selection	
Role identification and Responsibility Allocation	The project has poor project management plans for identifying roles and assigning the responsibilities	Identify few roles in the project management where some of the roles are left alone with any project responsibilities	Identify most of the roles in the project management and assign their responsibilities	Well planned project with proper role identification and responsibility allocation in the project management activities	
Impact identification					
Formatting and Submission	Project report is not complete and Several errors in spelling and grammar. Present a Confusing	Some errors in spelling and grammar. Some problems	Few errors in spelling and grammar. Presents most of the details in	Project report is complete and No errors in spelling and grammar. Consistently	

	organization of concepts, supporting arguments, and real-life example. Sentences rambling, and details are repeated.	of organizing the answer in a logical order of defining, elaborating, and providing real-life examples.	a logical flow of organization in definition, details, and example.	presents a logical and effective organization of definition, details, and real-life example of the topic.	
Acquired marks:					
CO Pass / Fail:					

Rubric for Project Assessment (CO4)

Marks Distribution (Maximum 3X5=15)					
Marking Criteria	Inadequate (1-2)	Satisfactory (3)	Good (4)	Excellent (5)	Acquired Marks
Project Planning	No background information regarding the project is given; project goals and benefits are missing.	Insufficient background information is given; project goals and benefits are poorly stated	Sufficient background information is given; the purpose and goals of the project are explained.	Thorough and relevant background information is given; project goals are clear and easy to identify.	
Effort Estimation and Scheduling	Student vaguely discuss the impact of societal, health, safety, legal and cultural issues in their project	Student provided with partial relevance to the impact of societal, health, safety, legal and cultural issues in their project	Student fairly provided the analysis to the impact of societal, health, safety, legal and cultural issues in their project	Student comprehensively provided the analysis to the impact of societal, health, safety, legal and cultural issues in their project	
Risk Management	Ambiguous representative example.	Partially identify / indicate towards real-life example.	Real-life example is fairly connected towards the definition.	Comprehensively defend with real life example.	
Acquired Marks:					
CO Pass / Fail:					